Syllabus Year: 2024-2025

CHRIST (Deemed to be University), Bengaluru - 560029

END-SEM EXAMINATIONS - APRIL 2025

Bachelor of Computer Applications SEMESTER - IV

Course Code: BCA302-4 **Max Marks: 50**

Course Name: PRINCIPLES OF SOFTWARE DEVELOPMENT - II **Duration: 02:00**

General Instructions:

- All rough work should be done in the answer script. Do not write or scribble in the question paper except your register number.
- Verify the Course code / course title and number of pages of questions in the question paper.
- Make sure your mobile phone is switched off and placed at the designated place.
- Malpractices will be viewed very seriously.
- Answers should be written on both sides of the paper in the answer book. No sheets should be detached from the answer book.
- Answers without the question numbers clearly indicated will not be valued. No page should be left blank in the middle of the answer book.

Course Outcome: Students will be able to:

CO1: Design and Develop ethical software product.

CO2: Apply terms, concepts, and tools of relational database management systems and PL/SQL concepts.

CO3: Construct database projects using DDL and DML queries.

CO4: Demonstrate the performance tune of Document-oriented NoSQL database.

CO5: Apply Nosql development tools on different types of NoSQL Database.

Section A **2*5=10**

Answer all the Questions

1)	Identify the advantages of SOA in modern enterprise systems.	**2**	(CO1, L2)
2)	Determine the features of B+ trees and describe their insertion and deletion implementation process.	**2**	(CO2, L3)
3)	List any four coding standards followed in popular software languages.	**2**	(CO3, L3)
4)	Detail the key features of Release Testing.	**2**	(CO4, L4)
5)	Differentiate Continuous Integration and Continuous Deployment.	**2**	(CO5, L2)

Section B **5*4=20**

6)	Describe the key principles of Service Oriented Architecture (SOA) and provide examples of technologies or standards commonly used in SOA implementation.	**5**	(CO1, L1)
7) a)	Consider the relation R(Student_ID, Course_ID,Instructor_Name, Grade, Credits) with the following functional dependencies: 1. Student_ID, Course_ID → Grade 2. Course_ID → Credits 3. Instructor_Name → Course_ID • What are the candidate keys for R? • What is the highest normal form that R satisfies? Justify your answer. • Decompose R into BCNF tables.	**5**	(CO2, L3)

OR

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b)	Design patterns have had a significant impact on object-oriented software design. Explain the importance of design patterns in software development with examples.	**5**	(CO2, L3)
8)	Detail safety and security issues in Software testing.	**5**	(CO4, L6)
9)	What is Node.js? how does it differ from traditional server-side technologies?	**5**	(CO5, L1)

Section C **10*2=20**

Answer all the Questions

10) a)	Explain different types of NoSQL databases and Discuss the challenges you might face when using multiple NoSQL databases together. How would you ensure data consistency and integrity across different databases?	**10**	(CO4, L5)
	OR		
b)	You are the Scrum Master for a team developing a mobile app. A key stakeholder, the marketing director, constantly requests changes to the app's UI/UX, often late in the sprint. These changes disrupt the team's focus, impacting sprint goals and velocity. The development team is becoming frustrated, and the Product Owner is struggling to balance the stakeholder's requests with the sprint backlog. a) How would you, as the Scrum Master, address this situation? b) What Scrum principles and practices would you leverage to resolve the conflict and ensure a productive working relationship between the stakeholder and the team? c) What steps would you take to prevent similar situations from occurring in future sprints?	**10**	(CO4, L5)
11)	Explain the efficiency and effectiveness of different types of loops (FOR, WHILE, and LOOP) in iterative processing for a given problem. Consider a scenario where you are required to process a list of student scores, calculate their average, and identify the highest and lowest scores. Compare the performance, readability, and error-handling capabilities of each loop type in the context of this problem. Write the PL/SQL code for each loop type to solve this problem, and justify which loop you would recommend based on your evaluation.	**10**	(CO3, L3)

Revised Bloom's Taxonomy (RBT) Levels		
L1 - Remember	L2 - Understand	L3 - Apply
L4 - Analyse	L5 - Evaluate	L6 - Create

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